

ABDALI, Kuwait

WMO: 405500

Lat: 30.067N

Lon: 47.683E

Elev: 83

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 37.3	(c) 40.4	(d) 13.0	(e) 10.7	(f) 55.7	(g) 17.1	(h) 13.0	(i) 57.3	(j) 20.0	(k) 53.8	(l) 16.6	(m) 57.5	(n) 6.1	(o) 320	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.4	119.3	70.9	117.4	70.6	115.6	70.2	81.6	95.0	78.4	95.6	75.6	97.8	10.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	146.4	89.8	73.8	126.4	87.6	69.6	109.1	86.5	45.6	93.0	42.1	96.0	39.1	96.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.2	(b) 17.8	(c) 15.3	(e) 31.6	(f) 123.0	(g) 3.5	(h) 1.5	(i) 29.2	(j) 124.1	(k) 27.1	(l) 125.0	(m) 25.2	(n) 125.8	(o) 22.7	(p) 126.9		
(a) 21.2	(b) 17.8	(c) 15.3	(e) 27.8	(f) 84.6	(g) 4.0	(h) 2.1	(i) 24.9	(j) 86.1	(k) 22.6	(l) 87.4	(m) 20.3	(n) 88.5	(o) 17.4	(p) 90.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	55.1	60.3	69.2	79.3	91.5	99.3	102.3	101.3	94.5	84.1	67.8	58.6
	DBStd	17.65	5.75	6.06	6.26	5.92	5.22	3.89	2.98	2.95	3.88	5.47	6.38	5.40
	HDD50	31	21	5	0	0	0	0	0	0	0	0	0	5
	HDD65	730	307	149	26	0	0	0	0	0	0	0	41	207
	CDD50	11122	180	294	594	879	1288	1480	1621	1590	1336	1057	533	270
	CDD65	6347	1	18	156	429	823	1030	1156	1125	886	592	124	7
	CDH74	102177	8	152	1590	5101	12823	18231	21035	20170	14446	7492	1073	56
	CDH80	73311	0	23	593	2709	8748	13921	16571	15708	10298	4381	355	4
Wind	WSAvg	6.6	5.3	6.2	6.4	6.6	6.8	9.1	9.0	7.2	6.4	5.4	5.0	5.3
Precipitation	PrecAvg	12.4	2.6	1.7	1.9	1.1	0.3	0.0	0.0	0.0	0.0	0.5	1.8	2.5
	PrecMax	17.8	6.5	4.0	5.9	4.0	1.9	0.1	0.1	0.0	0.3	2.2	7.6	6.8
	PrecMin	6.7	0.5	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
	PrecStd	3.2	1.5	1.0	1.3	0.9	0.4	0.0	0.0	0.0	0.0	0.6	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.1	83.2	99.7	105.1	114.7	120.5	122.2	120.5	116.2	108.0	92.4	79.3
		MCWB	56.6	59.3	62.0	65.7	68.9	69.8	70.9	71.8	70.6	68.5	63.0	57.1
	2%	DB	71.5	78.4	90.8	99.4	111.0	117.7	119.8	118.8	112.7	103.9	87.8	75.1
		MCWB	56.1	57.7	60.8	64.2	68.9	70.2	70.6	71.3	70.4	67.6	62.2	58.7
	5%	DB	68.7	75.0	86.0	96.2	108.2	115.5	117.6	116.9	110.4	101.2	84.0	72.3
		MCWB	55.7	56.7	59.4	63.1	67.5	69.3	70.7	70.9	69.4	67.2	62.0	57.8
	10%	DB	66.1	72.0	82.0	92.8	105.2	113.2	115.6	114.8	108.1	97.9	80.3	69.4
		MCWB	54.5	55.4	58.7	62.5	66.7	68.5	70.1	70.9	68.5	66.7	61.1	57.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	64.9	68.0	70.8	74.9	79.3	81.8	85.1	83.1	81.1	73.6	67.0
		MCDB	68.1	71.7	79.9	84.1	93.5	97.9	95.5	95.0	90.9	91.7	80.1	71.1
	2%	WB	61.1	62.6	66.3	68.7	72.0	72.9	77.6	82.8	80.0	77.3	69.5	64.7
		MCDB	66.0	71.4	77.2	85.8	96.8	111.7	100.6	95.3	94.6	85.2	78.1	69.3
	5%	WB	59.3	60.3	64.4	67.2	70.5	71.1	74.0	79.1	76.0	75.2	66.7	61.8
		MCDB	64.8	68.6	76.2	85.6	96.4	112.6	108.1	99.6	98.8	85.4	76.6	67.4
	10%	WB	56.8	58.2	62.2	65.7	69.1	69.5	72.1	75.3	72.3	72.6	64.4	59.2
		MCDB	63.5	67.3	75.8	83.9	95.9	110.7	111.3	105.0	102.5	85.1	73.8	65.6
Mean Daily Temperature Range	5% DB	MDBR	20.9	22.3	24.7	24.8	25.4	27.0	27.4	28.8	29.0	26.2	22.8	21.8
		MCDBR	22.8	25.0	30.5	28.6	28.1	29.1	29.5	30.2	30.0	29.3	26.5	24.1
		MCWBR	11.1	10.0	10.6	9.6	9.4	9.8	10.4	10.2	10.5	11.4	11.7	11.5
	5% WB	MCDBR	18.6	18.9	21.5	22.8	25.2	28.1	28.7	27.5	26.3	23.3	19.7	17.4
	MCWBR	10.5	9.6	9.7	8.3	8.6	10.8	11.8	13.6	12.9	11.9	11.1	10.6	
Clear-Sky Solar Irradiance	taub	0.500	0.542	0.600	0.692	0.690	0.632	0.666	0.615	0.560	0.579	0.539	0.503	
	taud	1.814	1.726	1.604	1.490	1.495	1.563	1.530	1.619	1.715	1.690	1.761	1.818	
	Ebn at Noon	220	225	222	209	211	222	214	223	230	211	206	211	
	Edh at Noon	57	68	81	94	94	88	90	82	72	69	59	54	
All-Sky Solar Radiation	RadAvg	1096	1389	1761	1932	2234	2515	2395	2261	1994	1555	1157	1014	
	RadStd	75	82	69	101	103	75	66	53	65	66	87	91	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.52	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	+153	+149

Nomenclature: See separate page